

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I __G. Ram Charan (192525043)__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.¹⁹

Signature of the student: _____G. Ram Charan_____

Criteria	Technical Content (1.5)	Structure and Flow (1)	Use of Visuals (0.75)	Communication Skills (1)	Professionalism (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

1. A cloud provider aims to improve data center efficiency. Present how hardware evolution (CPU virtualization support, GPUs, SSDs, high-speed networking) enhances cloud performance and scalability.
2. A company is redesigning its application using microservices and APIs. Present how the evolution from traditional web applications to cloud-native software has influenced modern application design.
3. A data center needs to maximize resource utilization. Prepare a technical presentation on server virtualization, covering hypervisors, virtual machines, and containerization. Compare their roles in cloud environments.
4. An enterprise integrates multiple distributed systems. Present how web services enable interoperability in cloud platforms, including standards, protocols, and service-oriented architecture (SOA).
5. An educational institution plans to adopt cloud services for labs, development, and administration. Present a comparative analysis of IaaS, PaaS, and SaaS, focusing on control, flexibility, cost, and use cases.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

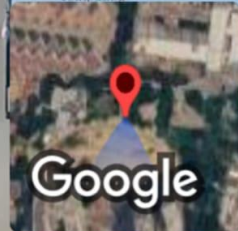
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

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Introduction & Problem Statement

- **Legacy IT Crisis:** On-premise campus data centers suffer from high capital costs (CapEx) and low hardware utilization.
- **Enrollment Bottlenecks:** Seasonal traffic spikes during registration week routinely crash legacy server infrastructure.
- **Administrative Burden:** IT staff spend excessive time patching physical servers instead of enabling academic innovation.
- **The Cloud Solution:** Cloud service tiers decouple hardware from software, delivering elastic, pay-as-you-go computing.



institutional workloads (Labs, Dev, Admin) to the optimal cloud service model

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